THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 9: **Cell: The Building Block of Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Osmosis is a special case of diffusion. Reason (R): It involves the movement of water across a selectively permeable membrane. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. The ratio of the number of chromosomes in a human zygote to that in a sperm is: 1

(a) 1 : 2 **(b)** 1 : 1 **(c)** 4 : 1 **(d)** 2 : 1

3. A wilted lettuce is placed in water. In an hour it becomes crisp due to: 1

(a) turgor in cells **(b)** cell division **(c)** photosynthesis **(d)** respiration

4. The organelle that stores starch granules in the potato is: 1

(a) chloroplast **(b)** leucoplast **(c)** chromoplast **(d)** lysosome

5. Assertion (A): Bacteria are prokaryotes. Reason (R): They do not have a well-defined nucleus enclosed by a membrane. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. The surface area to volume ratio of a cube of side 2 cm compared with a cube of side 1 cm is: 1

(a) half **(b)** double **(c)** same **(d)** four times

7. A beetroot piece placed in cold water stays coloured but in hot water the water becomes red because: 1

(a) heat denatures the membrane proteins **(b)** colour evaporates
(c) water boils **(d)** cells divide

8. A prokaryotic cell has ribosomes of smaller size and: 1

(a) no membrane-bound organelles **(b)** a nucleus
(c) ER **(d)** mitochondria

9. Assertion (A): Cell size is not related to the size of the organism. Reason (R): Larger organisms have more cells, not larger cells. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. If a cell's diameter doubles, its volume increases by: 1

(a) 8 times **(b)** 4 times **(c)** 2 times **(d)** 16 times

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. State two uses of tissue culture in agriculture. 2

12. Why are the cells of the root cap and meristem not rich in chloroplasts? 2

13. State two differences between osmosis and active transport. 2

14. Why does the cytoplasm contain many organelles? Give two reasons. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe the endomembrane system and the pathway of a secreted protein. 3

16. A cell is placed in a solution and no change in size occurs but later the mass decreases. Suggest the possible explanation (solutes entering or the solution concentration changing). 3

17. Explain the importance of the water balance in cells for the preservation of fruits by sugar (jam) and salting. 3

OR

Why are viruses not considered cells? Compare with bacteria in three points. 3

18. Why are cells of the epidermis in plants flat and those of the palisade tissue columnar? Relate to function. 3

OR

Compare diffusion rates in a small unicellular organism and a large multicellular one, showing why the latter needs transport systems. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Compare mitosis and meiosis in a table of at least five points and explain the significance of each for living organisms with the role of crossing over. 5

OR

Discuss the discovery of cell structures by different scientists (Hooke, Leeuwenhoek, Brown, Schleiden, Schwann, Virchow) and the role of technology like electron microscopes in revealing organelles. 5

20. Describe the cell organelles which are found in both plant and animal cells and those found in only one. Mention structure and function of each with a table of 8 organelles. 5

OR

Explain the role of the cell membrane in protection, transport and communication, and how it helps in keeping the internal environment of the cell constant. 5

END OF QUESTION PAPER — Total: 40 Marks